Yoshwan Pathipati

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Education

Virginia Tech Blacksburg, VA

B.S. in Computer Science; Minors in Artificial Intelligence, Cybersecurity, and Mathematics *Aug. 2023 – May 2027*

- **Honors:** Dean's List (top 20% of class) **Leadership:** Co-founder, Collaborative Coding Activities Club (80+ members)
- **Coursework:** Machine Learning, Intro to AI, Data Structures & Algorithms, Cloud Software Development, Cryptography

Technical Skills

Languages: Python, Java, C, JavaScript, TypeScript, SQL, Ruby, Bash, PowerShell, HTML/CSS
AI/ML: JAX, Ollama, Llama 3, RAG, LLM APIs, faster-whisper, Modern Hopfield Networks, Equilibrium Propagation
Cloud & DevOps: AWS (VPC, EC2, IAM, S3, GuardDuty), Azure, GCP, Terraform, Docker, GitHub Actions, Linux
Security: Nessus, OpenVAS, Wireshark, nmap, tfsec, NIST 800-53 / RMF, CMMC, OWASP LLM Top 10
Web & Data: React, Node.js, Express, Flask, Django, Ruby on Rails, Twilio, Stripe, PostgreSQL, MySQL, MongoDB, Redis
Certifications: AWS Certified Solutions Architect – Associate (Jul. 2026); CompTIA Security+ (in progress)

Experience

AI Research Assistant, LLM Automation Sep. 2026 – Present

Virginia Tech, Dept. of Engineering Education *Blacksburg, VA*

- Develop Python preprocessing and faculty-reviewed prompts that generate CATME feedback via Ollama for 100+ students
- Assess Llama, Qwen, and Kimi on VT compute, guiding faculty to models that meet classroom performance and cost goals
- Containerize the feedback pipeline with Docker alongside faculty, standardizing setup for a 100+ student cohort

Cybersecurity Intern May 2026 – Aug. 2026

Triple Point Security *Blacksburg, VA*

- Ran authenticated and unauthenticated Nessus scans on 5 hosts, finding 20+ critical and 84 high-severity issues
- Compared self-hosted Llama 3 with cloud LLMs for vulnerability-report generation, exposing severity-classification errors, missed CVEs, and RAG reliability limits that led to updated detection rules and fewer false positives
- Built an AWS–Azure site-to-site VPN, fixing an IKE policy mismatch; provisioned AWS, Azure, and GCP with Terraform
- Kept a stalled OpenVAS assessment on schedule by driving scans over GMP from the CLI in its Docker deployment
- Resolved a NIST/CMMC mismatch across 4 publications, setting the control structure for Security Hub compliance mapping
- Introduced EventBridge and SNS to the compliance team and trained 3 teammates to deploy workflows across AWS Regions

Research Assistant, Full-Stack Development (SpaceNet Testbed) Sep. 2025 – Present

Hume Center, National Security Institute & Dept. of Aerospace and Ocean Engineering *Blacksburg, VA*

- Lead development of SpaceNet, a satellite-constellation simulation testbed, building its Flask REST APIs for 5 researchers
- Replaced SSH config editing with a browser workflow and 4 React telemetry dashboards used by 10+ team members
- Dockerized the platform and automated CI/CD with GitHub Actions, cutting deployment time from 45 to 8 minutes
- Presented SpaceNet as a research poster at the Virginia Tech Summer Research Conference (2026)

Research Assistant, Backend Development (OpenDSA) Jun. 2025 – Aug. 2025

Virginia Tech, Dept. of Computer Science *Blacksburg, VA*

- Refactored 3 Rails controllers in OpenDSA's analytics modules, removing N+1 queries and cutting query time by 30%
- Built Rails features for CS 3114's learning system: automated grading and real-time feedback that reduced manual grading

Python Developer Intern Jul. 2024 – Oct. 2024

Avenues, IIT Bombay *Remote*

- Shipped Django and MySQL backend features (authentication, profiles, data validation) across three core modules
- Found 6 critical auth-flow bugs through API testing with Postman and the DRF test client, all fixed within two sprints

Projects

CloudGuard | *Terraform, AWS, GuardDuty, Inspector, Security Hub, React, GitHub Actions, tfsec* Jan. 2026 – Apr. 2026

- Terraform-built AWS (VPC, EC2, S3, IAM) running GuardDuty, Inspector, and Security Hub; findings map to NIST 800-53
- Live React dashboard of findings; tfsec in GitHub Actions blocks any merge that introduces a misconfiguration

WhisperKey | *Python, faster-whisper, Ollama, CUDA* github.com/YoshwanPathipati/whisperkey

- Open-source, offline Wispr Flow alternative: hold a key, speak, and formatted text lands at the cursor in any Windows app
- Local GPU speech-to-text (faster-whisper, CUDA) plus an Ollama LLM cleanup pass; no audio ever leaves the machine

PromptShield | *Python, LLM APIs, Ollama, OWASP Top 10 for LLMs* In progress

- LLM security harness firing injection, jailbreak, and data-exfiltration prompts mapped to the OWASP Top 10 for LLMs
- Building input/output guardrails to cut attack success rate; benchmarking self-hosted Llama 3 against a cloud model

AI Restaurant Phone Assistant | *Twilio, LLM API, Flask, React*

- Twilio and LLM phone agent that answers calls, handles FAQs, and books reservations, with a Flask backend for call state

Multimodal Associative Memory | *JAX, Modern Hopfield Networks, Equilibrium Propagation* Feb. 2026 – Present

- Continuous-time JAX dynamics and a Log-Sum-Exp Modern Hopfield core for bidirectional audio-visual recall

Organic Farm E-Commerce Platform | *React, Node.js, Express, MongoDB, Stripe* Oct. 2025 – Present

- Storefront for my family's farm with Stripe payments and real-time inventory for 100+ SKUs, replacing spreadsheet orders